

Power Function & Types of Error

CSU, Chico Math 351

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Hypothesis Testing, the decision

We conclude a formal hypothesis test by making a decision between H_0 and H_1 . But did we decide correctly?

Even if all our calculations are accurate, our decision could be correct or incorrect.

	fail to reject H_0	reject H_0
H_0 true	Correct	Type I Error
H_A true	Type II Error	Correct

Hypothesis Testing, errors

Type I Error

A Type I (1) Error is rejecting the null hypothesis when H_0 is actually true.

This occurs with probability α , which we have seen previously. It can only occur in cases where you **reject** H_0 .

Type II Error

A Type II (2) Error is failing to reject the null hypothesis when the alternative is actually true.

This occurs with probability β , which we will define in this section. It can only occur in cases where you **fail to reject** H_0 .

Probability of Type I Error: α

When choosing a value for α , we ask "how much evidence do we need to reject the null hypothesis?" Setting α too high means we have a higher likelihood of rejecting H_0 when we shouldn't.

It follows that a low α is good for reducing the probability of a false positive.

However...

Hypothesis Testing, error trade offs

Consider the example of a court system that runs on the premise of “innocent until proven guilty.” In hypothesis terms:

H_0 : Defendant is innocent; H_1 : Defendant is guilty

How could we reduce the Type II Error rate in court? What influence would this have on the Type I Error rate?

- ▶ Lower Type II Error by convicting more (guilty) people.
- ▶ What then of wrongful convictions, i.e. Type I Error rates?

A trade off exists when considering both types of error.

Probability of Type II Error: β

Unlike α , β is not something we can decide ahead of time, and it will change depending on the value for which we fail to reject H_0 . The value for β addresses: "what is the probability of failing to reject H_0 for a specific parameter value?"

This means β has many possible values, and can be described with a function dependent on the assumed value of the parameter.

There may be cases where both α and β are small (if the value of rejection is very far away from the parameter value). In any case, a decrease in α necessitates an increase in β .

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Example: Fuel Additive

An automobile company is testing additives that may increase gas mileage. They send 30 cars fueled by the new additive on a road trip, tracking their mileage. Without the additive, these cars average 25 mpg with a population standard deviation $\sigma = 2.4$. If we test at significance level $\alpha = 0.05$, this means we know:

$$\mu_0 = 25$$

$$\sigma = 2.4$$

$$\alpha = 0.05$$

$$n = 30$$

$$z_{1-\alpha}^* = qnorm(1 - \alpha) = 1.6449$$

Example: Fuel Additive

In this case, the decision rule says we **reject** H_0 for all test statistics such that

$$z > z_{1-\alpha}^*$$

$$\frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} > z_{1-\alpha}^*$$

$$\bar{X} > ((\sigma/\sqrt{n}) * z_{1-\alpha}^*) + \mu_0$$

$$\bar{X} > ((2.4/\sqrt{30}) * 1.6449) + 25$$

$$\bar{X} > 25.72074$$

By extension, we will **fail to reject** H_0 for $\bar{X} < 25.72074$.

Example: Fuel Additive

Since we **fail to reject** H_0 for $\bar{X} < 25.72074$, we can calculate β for our choice of true $\mu = \mu_A$. In the case where $\mu = 26$, for example:

$$\begin{aligned} &P(\bar{X} < 25.72074 | \mu = 26) \\ &= P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < \frac{25.72074 - \mu}{\sigma/\sqrt{n}}\right) \\ &= P\left(\frac{\bar{X} - 26}{2.4/\sqrt{30}} < \frac{25.72074 - 26}{2.4/\sqrt{30}}\right) \\ &= P(Z < -0.6373208) = \text{pnorm}(-0.6373208) = 0.2619 = \beta \end{aligned}$$

There is a probability of 0.2619 that we make a Type II error in this situation.

Question: What would be the Type II error for $\mu = 27$?

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Exercise: Sales Problem

Prospective sales people for an encyclopedia company are now being offered a sales training program. Previous data indicate that the average number of sales per month for those who do not participate in the program is 33.

Assume that $\sigma = 8.4$. To determine whether the training program is effective, a random sample of 35 new employees is given the sales training and then sent out into the field. One month later, the mean for the number of sets of encyclopedias sold was 35.

Do the data present sufficient evidence to indicate that the training program enhances sales? Use $\alpha = .05$.

Exercise: Sales Problem

- ▶ Establish hypotheses: null (H_0) and alternative (H_1).
- ▶ Calculate test statistics and p-value
- ▶ State conclusion (reject H_0 ? Context of problem?)
- ▶ Based on conclusion, if an error was made, was it Type I or Type II?

Exercise: Sales Problem

Suppose that the company thinks that the cost of financing the sales program will be offset by increased sales if those in the program average 38 sales per month. Compute β for $\mu = 38$ and, based on the value of β , indicate whether the power of the test is reasonable here.

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References

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